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AI for Vision & NLP

CA2 - Application

Assignment:	AI for Vision and NLP – CA2: Application
Learning Outcomes:	▪ Differentiate between Text Analytics and NLP systems. ▪ Critique, analyse and examine key components required to successfully build a text analytics system. ▪ Design and implement a text analytics and NLP system to create meaningful output suited to a particular use case. ▪ Analyse the various elements of a Computer Vision system. ▪ Design, develop and implement algorithmic techniques to build a Computer Vision system. ▪ Critically examine case studies to identify a Computer Vision problem and implement a suitable solution.
Due Date:	Final Application and Code : Tuesday **12th May 2026** at **midnight** Recorded Presentation : Tuesday **12th May 2026** at **midnight**
Submission Instructions:	The application, code and data/dataset should be submitted to Blackboard on the link provided. The recorded presentation should be submitted to Blackboard on the link provided.

Notes

This is an individual assignment and there should be no **plagiarism of online resources or malpractice. Ensure to cite online resources if using them.** Students have a responsibility to keep their own work secure and maintain the integrity and validity of their academic output.

Task

Develop a multi-modal document understanding system that integrates text and image processing to analyse and extract meaningful insights from scanned or photographed documents. The documents may contain both textual and visual elements, such as forms, tables, images, or diagrams.

This assignment is worth a total of 40% of the module grade for this module. The complexity of the application will have a direct impact on the marks received.

The system will:

1. Extract text from images (using OCR).
2. Process and analyse the extracted text (tokenisation, stemming, lemmatisation, stopword removal).
3. Detect and recognise key features in both the text and images (e.g., tables, signatures, diagrams, figures).
4. Provide a structured output with textual and visual insights, such as categorising sections, extracting tabular data, recognising diagrams, and detecting important patterns or key phrases.

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Application Breakdown

1. Text Extraction and Processing
   - OCR Text Extraction: Use Tesseract OCR to extract text from a diverse set of document images (e.g., scanned documents, photos of printed forms, PDFs, etc.)
   - Text Preprocessing: removing stop words, performing tokenisation, and applying stemming or lemmatisation.
   - Text Feature Extraction: Extract meaningful features using TF-IDF or Bag-of-Words, and extract key phrases, named entity recognition (NER), or even sentiment analysis
   - Text Analysis: Categorising sections (e.g., separating titles, body text, and captions), Extracting tabular data, Analysing the relationships between text and images, such as identifying references to figures, diagrams, or tables in the text

2. Image Processing and Feature Detection
   - Image Preprocessing: Use OpenCV to pre-process the document(grayscale, thresholding, noise reduction, or other techniques to enhance the quality of the images)
   - Feature Detection: Detect key visual features (e.g., text blocks, tables, images, and diagrams) in the document using methods like edge detection, feature matching (e.g., SIFT, ORB), and contour detection.
   - Image Segmentation: Segment the document into meaningful regions, such as isolating text areas, separating images, diagrams, and tables using OpenCV's image segmentation techniques like watershed or connected components.
   - Image Enhancement: Image refinement (contrast adjustment, deblurring) and text enhancement (binarisation) to improve the legibility and overall quality of the extracted text and features.

3. Multi-modal Integration and Insights
   - Text and Image Integration: Integrate both the text and image processing components to produce a unified understanding of the document
   - Final Output: Create a final output that presents
     - The processed text, including cleaned text, feature extraction results, and analyses.
     - The refined image, showing both the original and the processed version (e.g., image with segmented parts and detected features).
     - Insights or reports, such as extracting specific data, categorising document sections, or visualising detected features.

This assignment is intentionally a bit vague and open-ended; I am looking for you to show initiative and inventiveness.

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Deliverables

1. Python notebook/code: A fully functional system that implements the above tasks, with clear organisation and comments
   - A detailed Python notebook that:
     - Explains each step of the process, from OCR extraction to text and image analysis.
     - Describes the techniques and algorithms used, along with any challenges faced and how they were overcome.
     - Includes visualisations and examples of the extracted and processed text, as well as images showing the results of feature detection and segmentation.
   - The code must use meaningful headings and be structured in a neat and meaningful manner. Any code taken from online must be referenced.
   - The data/dataset used must be uploaded as part of the submission.
   - All data used must be uploaded as part of the submission
   - The code must be fully executed and the output should be easily seen in both code and presentation – this is the students responsibility, especially in the presentation.

2. Recorded Presentation: 10min recorded presentation that demonstrates the working system
   - No powerpoint - code and working system only.
   - Students will get a mark of zero if the system is not demonstrated.
   - Discuss the challenges encountered in the project and how they were addressed.
   - Presents examples of documents that were processed and shows the output (both text and images).
   - The student must demonstrate knowledge and understanding of the computer vision techniques and code within their application.
   - The presentation will be a strict 10min. The challenge is to demonstrate the application in the best light and your knowledge within this tight/strict timeline.
   - When recording the students must have an insert of themselves so they can be seen talking for the whole presentation – OBS Studio or something similar can be used.
   - The student must check the recording before submitting – ensure that the student can be seen and heard and there is no background noise. If the student cannot be herd or seen then the submission will receive a mark of zero

3. GitHub account : using GitHub Classroom for weekly submission
   - Students must submit weekly – pushing your code submits your code
   - Do not rename or delete your repository

<u>Instructions</u>

- Make sure you are logged into your GitHub account.
- Click the assignment link I send you (likely email)
- When prompted, accept the assignment.
- GitHub will automatically create a private repository for you.
- Open the repository and start working on the files.
- When you make changes, commit and push them to GitHub.
- I will see your work automatically — no need to upload anything elsewhere.

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Further Guidance

The code must be commented and a description provided, use meaningful headings and be structured in a neat and meaningful manner. Any use of online resources must be referenced in the code. If not referenced then the a student may have to attend a verification of assessment as per 4.5.4 Verification of Assessment of the Marks and Standards Policy.

Late submissions will have a penalty applied as per 4.11 Extension to a Deadline in the Marks and Standards Policy.
Students must apply the principles and standards of the Academic Integrity Policy and Marks and Standard Policy when submitting.